

LINUX ASSIGNMENT

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Command line arguments, fileoperators, read and filters

Write a Bash script that accepts a filename as the first command-line argument.

Check whether an argument is provided.

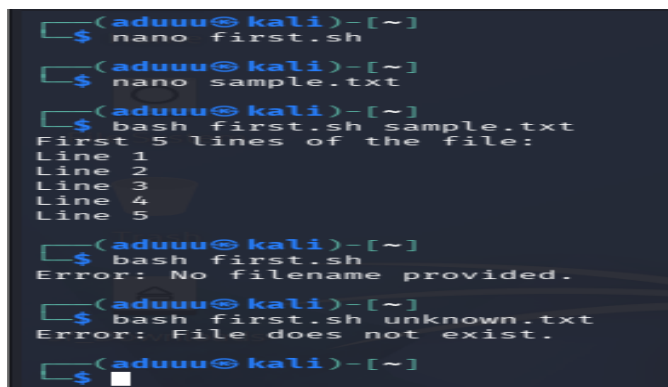
Check whether the file exists using a file operator.

If the file exists, display the first 5 lines using head.

Otherwise, display an appropriate error message.

File Type and Permission Checker

```
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No filename provided."
    exit 1
fi
if [ -e "$1" ]
then
    echo "First 5 lines of the file:"
    head -n 5 "$1"
else
    echo "Error: File does not exist."
fi
```



```
(aduuu@kali)-[~]
$ nano first.sh
(aduuu@kali)-[~]
$ nano sample.txt
(aduuu@kali)-[~]
$ bash first.sh sample.txt
First 5 lines of the file:
Line 1
Line 2
Line 3
Line 4
Line 5
(aduuu@kali)-[~]
$ bash first.sh
Error: No filename provided.
(aduuu@kali)-[~]
$ bash first.sh unknown.txt
Error: File does not exist.
(aduuu@kali)-[~]
$
```

Write a Bash script that accepts a path as a command-line argument and determines whether it is:

a regular file,

a directory,

readable,

writable, and

executable.

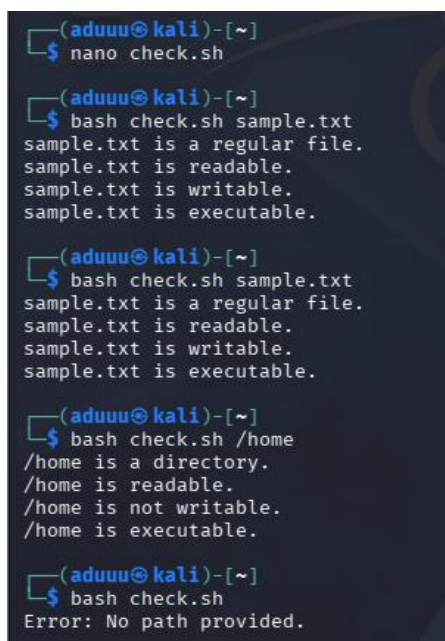
Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

```
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No path provided."
    exit 1
fi
if [ -f "$1" ]
then
    echo "$1 is a regular file."
elif [ -d "$1" ]
then
    echo "$1 is a directory."
else
    echo "$1 is neither a regular file nor a directory."
fi
if [ -r "$1" ]
then
    echo "$1 is readable."
else
    echo "$1 is not readable."
fi
if [ -w "$1" ]
then
    echo "$1 is writable."
```

```

else
    echo "$1 is not writable."
fi
if [ -x "$1" ]
then
    echo "$1 is executable."
else
    echo "$1 is not executable."
fi

```



```

(aduuiu@kali)-[~]
$ nano check.sh

(aduuiu@kali)-[~]
$ bash check.sh sample.txt
sample.txt is a regular file.
sample.txt is readable.
sample.txt is writable.
sample.txt is executable.

(aduuiu@kali)-[~]
$ bash check.sh sample.txt
sample.txt is a regular file.
sample.txt is readable.
sample.txt is writable.
sample.txt is executable.

(aduuiu@kali)-[~]
$ bash check.sh /home
/home is a directory.
/home is readable.
/home is not writable.
/home is executable.

(aduuiu@kali)-[~]
$ bash check.sh
Error: No path provided.

```

Write a Bash script that accepts a log file as a command-line argument and searches for the words: failed, denied, unauthorized, and error.

```

#!/bin/bash

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

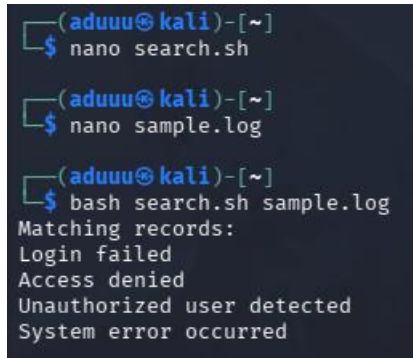
if [ ! -f "$1" ]
then

```

```
    echo "Error: Log file does not exist."
    exit 1
fi

echo "Matching records:"

grep -iE "failed|denied|unauthorized|error" "$1"
```

A terminal window on a Kali Linux system. The prompt is (aduuu@kali)~. The user runs 'nano search.sh', then 'nano sample.log', and finally 'bash search.sh sample.log'. The script outputs 'Matching records:' followed by five lines of log data: 'Login failed', 'Access denied', 'Unauthorized user detected', and 'System error occurred'.

Write a Bash script that accepts a filename and checks whether:

- the file exists,**
- it is a regular file,**
- it is empty or non-empty.**

If it is non-empty, display the first 5 lines.

```
#!/bin/bash

if [ -z "$1" ]
then
    echo "Error: No filename provided."
    exit 1
fi

if [ ! -e "$1" ]
then
    echo "Error: File does not exist."
    exit 1
fi
```

```

echo "File exists."
if [ -f "$1" ]
then
    echo "It is a regular file."
else
    echo "It is not a regular file."
    exit 1
fi
if [ -s "$1" ]
then
    echo "File is non-empty."
    echo "First 5 lines:"
    head -n 5 "$1"
else
    echo "File is empty."
fi

```

```

(adiuu@kali)-[~]
└─$ nano filecheck.sh

(adiuu@kali)-[~]
└─$ bash filecheck.sh sample.txt
File exists.
It is a regular file.
File is non-empty.
First 5 lines:
Line 1
Line 2
Line 3
Line 4
Line 5

```

```

(adiuu@kali)-[~]
└─$ touch empty.txt

(adiuu@kali)-[~]
└─$ bash filecheck.sh empty.txt
File exists.
It is a regular file.
File is empty.

(adiuu@kali)-[~]
└─$ ls -l empty.txt
-rw-rw-r-- 1 aduuu aduuu 0 Sep  9 00:00 empty.txt

(adiuu@kali)-[~]
└─$ bash filecheck.sh /home/adiuu/empty.txt
File exists.
It is a regular file.
File is empty.

(adiuu@kali)-[~]
└─$ nano empty.txt

(adiuu@kali)-[~]
└─$ bash filecheck.sh empty.txt
File exists.
It is a regular file.
File is empty.

```

Explanation of the operators:

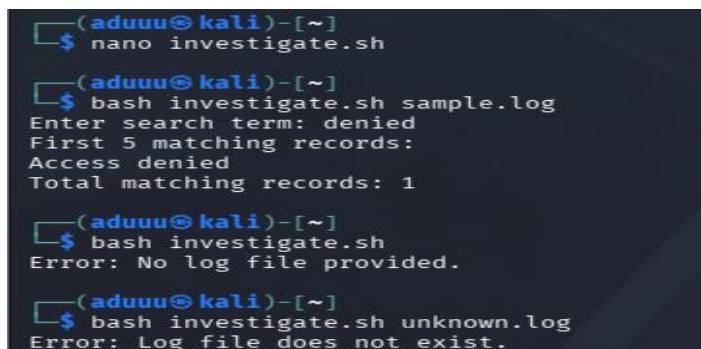
- e Checks whether the file exists
- f Checks whether the path is a regular file
- s Checks whether the file is non-empty
- head -n 5 Displays the first five lines

Write a Bash script named `investigate.sh` that accepts a log file as a command-line argument.

The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.

Display the first 5 matching records and report the total number of matching records.

```
#!/bin/bash
if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi
if [ ! -e "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi
if [ ! -r "$1" ]
then
    echo "Error: Log file is not readable."
    exit 1
fi
read -p "Enter search term: " term
echo "First 5 matching records:"
grep -i "$term" "$1" | head -n 5
count=$(grep -ic "$term" "$1")
echo "Total matching records: $count"
```



```
(aduuu@kali)-[~]
$ nano investigate.sh

(aduuu@kali)-[~]
$ bash investigate.sh sample.log
Enter search term: denied
First 5 matching records:
Access denied
Total matching records: 1

(aduuu@kali)-[~]
$ bash investigate.sh
Error: No log file provided.

(aduuu@kali)-[~]
$ bash investigate.sh unknown.log
Error: Log file does not exist.
```